



DIGITAL TWIN/BEER GAME ON STEROIDS

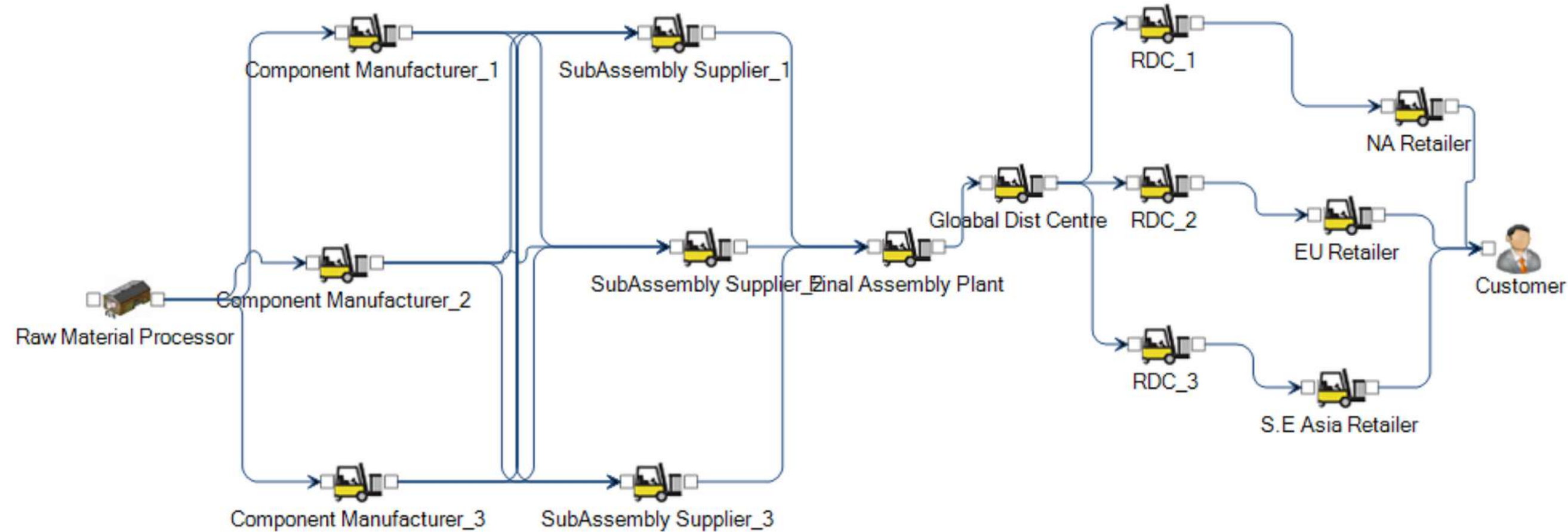
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CASE BACKGROUND: 16-ENTITY SUPPLY CHAIN



Customer Orders: 300 per week Initial and up until week 120

America Retailer: 100% from RDC 1

European Retailer: 100% from RDC 2

E. Asian Retailer: 100% from RDC 3

RDC 1, 2 and 3: 100% from Global Distribution Center

Global Distribution Centre: 100% from final assembler

Final Assembler: 70% Subassembly 1, 20% Subassembly 2, 10% Subassembly 3

Subassembly Supplier 1: 65% Component 1, 35% Component 2

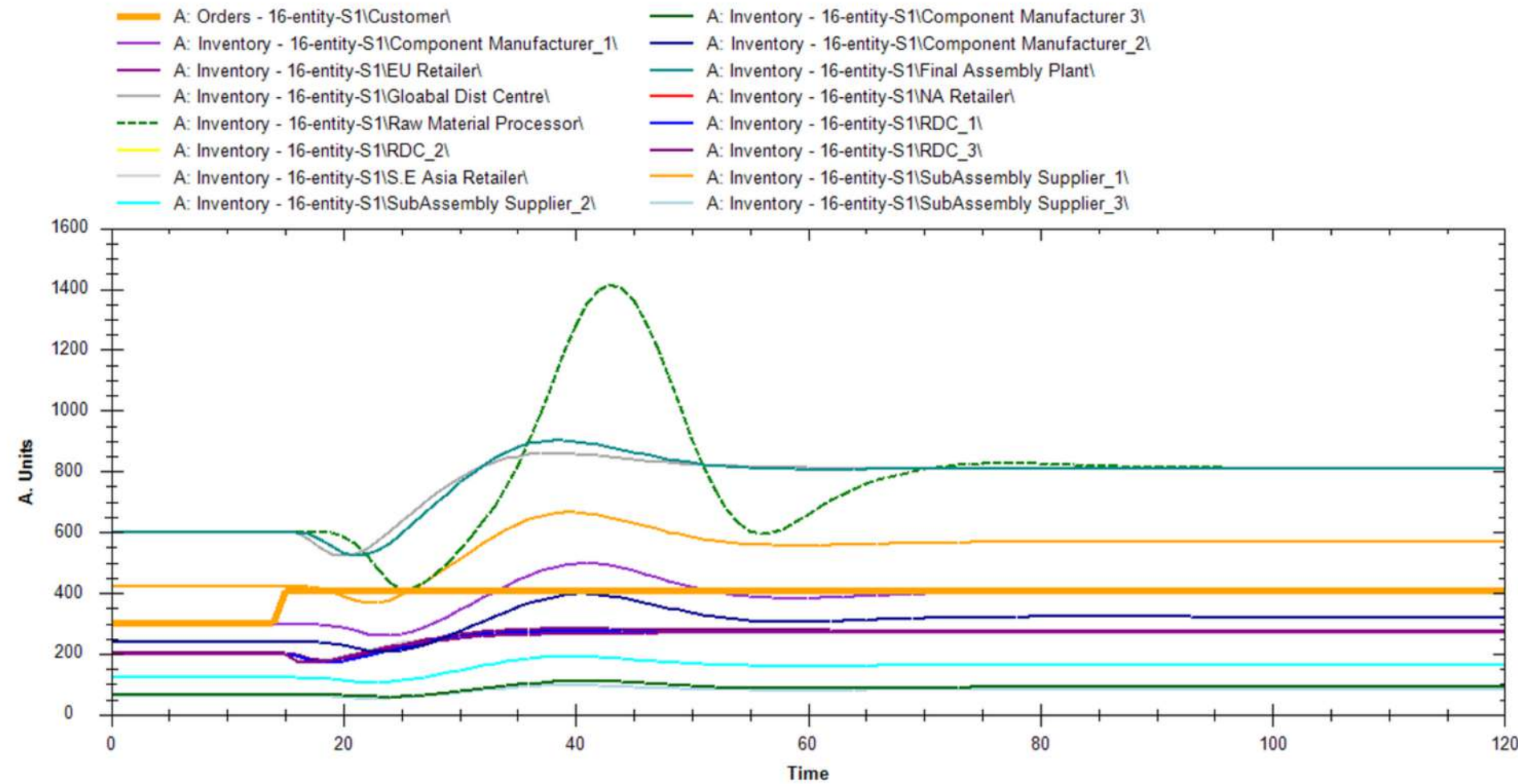
Subassembly Supplier 2: 55% component 2, 45% component 3

Subassembly supplier 3: 40% component 1, 40% component 2, 20% component 3

Component Manufacturers 1,2 and 3: 100% Raw Material Processor



SCENARIO 1: AVOID THE BULLWHIP EFFECT



Initial customer order: 300 units per week
Week 15: Orders increase by 35%
New order rate: 405 units per week (300×1.35)
Duration: Sustained until Week 120

WEEK	10	20	30	40
Component Manufacturer 1	297	285.78	343.06	497.77
Component Manufacturer 2	237	228.05	273.75	397.21
Component Manufacturer 3	66	63.51	76.23	110.61
Customer	300	405	405	405
European Retailer	198	188.76	250.92	265.07
Final Assembly Plant	600	532.94	761.95	898.63
Global Dist. Centre	600	521.63	777.02	857.02
N.America Retailer	198	188.76	250.92	265.07
Raw Material Processor	600	586.2	540.83	1,280.30
RDC 1	198	176.25	255.54	272.6
RDC 2	198	176.25	255.54	272.6
RDC 3	204	181.59	263.28	280.86
S.E. Asia Retailer	204	194.48	258.52	273.1
SubAssembly Supplier 1	420	389.42	511.46	665.23
SubAssembly Supplier 2	120	111.26	146.13	190.07
SubAssembly Supplier 3	60	55.63	73.07	95.03

1. Smooth Demand Increases

Instead of a sudden 35% increase in week 15, gradually raise customer orders over several weeks (e.g., 10% per week). This reduces panic upstream and allows each supply chain tier to adjust incrementally.

Trade-off: Slower initial ramp-up but avoids large inventory swings.

2. Improve Forecasting and Information Sharing

Ensure real-time demand data is shared across all entities—retailers, RDCs, manufacturers, and suppliers. This reduces uncertainty and prevents overproduction.

Trade-off: Requires investment in digital systems and coordination.

3. Reduce Lead Times and Adjust Safety Stock Dynamically

Shorten production and delivery lead times (e.g., from 4 weeks to 2-3 weeks) and use variable safety stock levels based on demand trends.

Trade-off: May increase operational pressure and require more responsive logistics.

4. Use a Pull-Based Replenishment System

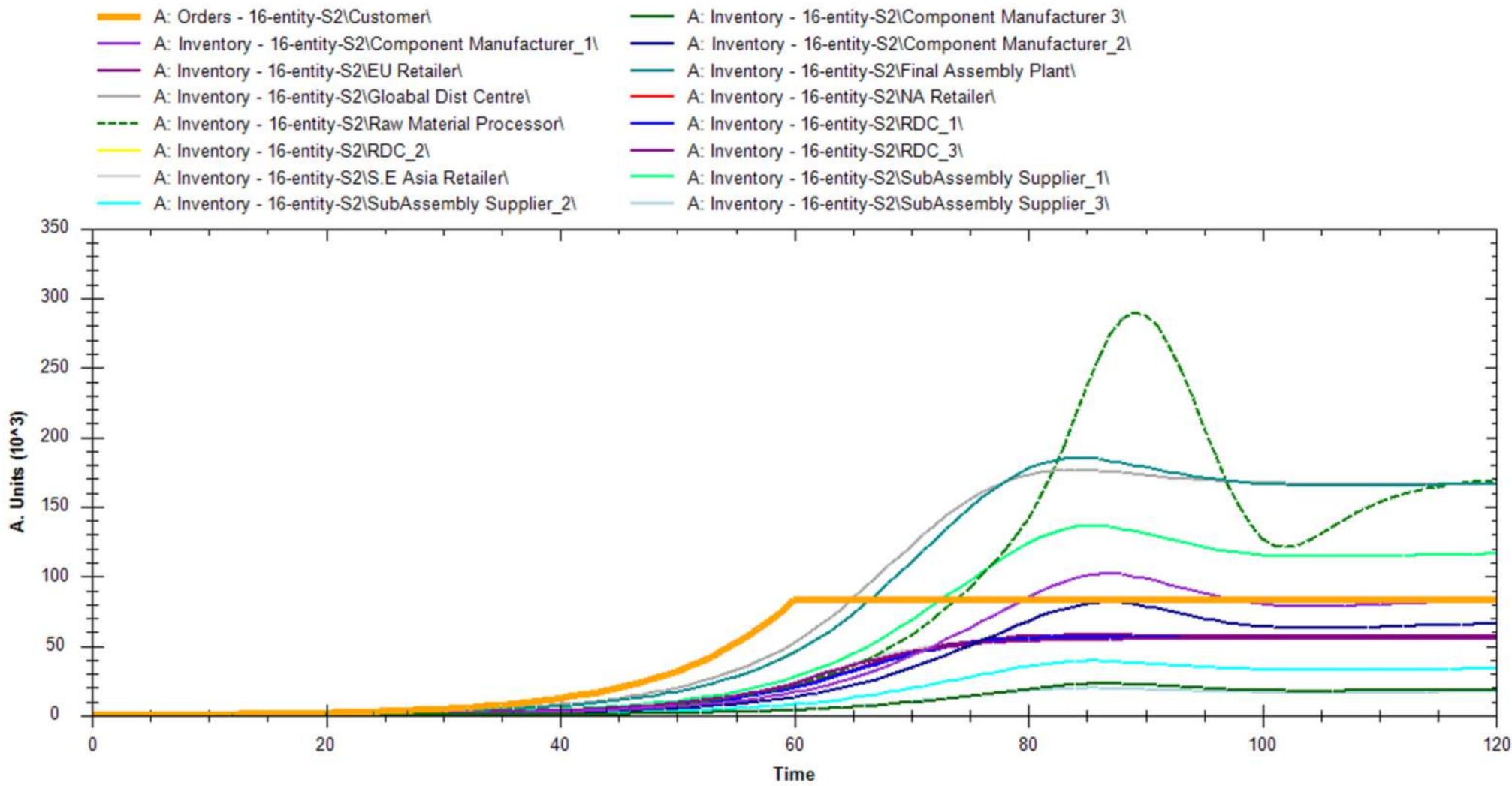
Base orders on actual consumption rather than forecasts alone. Set limits on how much upstream suppliers can increase production week-to-week.

Trade-off: Limits flexibility for sudden demand spikes but avoids overstock.



SCENARIO 2 : ASSUME UNKNOWN SALES INCREASE EACH WEEK

Initial order volume: 300 units per week
Starting Week 2: Demand increases by 10% each week
Growth Pattern: Compounding increase (week-over-week)
Duration: Continues until Week 60



WEEK	10	20	30	40	50	60
Component Manufacturer 1	264.82	414.82	974.11	2,483.54	6,429.80	16,674.83
Component Manufacturer 2	211.32	331.02	777.32	1,981.81	5,130.85	13,306.18
Component Manufacturer 3	58.85	92.18	216.47	551.9	1,428.84	3,705.52
Customer	707	1,835.00	4,759.00	12,343.00	32,016.00	83,040.00
European Retailer	204.96	512.37	1,327.70	3,443.86	8,932.40	23,168.47
Final Assembly Plant	518.23	1,039.20	2,601.90	6,726.01	17,441.88	45,239.32
Global Dist. Centre	533.69	1,176.47	3,002.30	7,778.91	20,175.65	52,330.44
N.America Retailer	204.96	512.37	1,327.70	3,443.86	8,932.40	23,168.47
Raw Material Processor	517.84	615.71	1,342.73	3,331.73	8,584.90	22,251.55
RDC 1	187.18	443.89	1,144.99	2,969.24	7,701.40	19,975.51
RDC 2	187.18	443.89	1,144.99	2,969.24	7,701.40	19,975.51
RDC 3	192.85	457.34	1,179.69	3,059.22	7,934.77	20,580.83
S.E. Asia Retailer	211.17	527.89	1,367.93	3,548.21	9,203.08	23,870.54
SubAssembly Supplier 1	363.68	650.1	1,583.06	4,072.18	10,554.73	27,375.01
SubAssembly Supplier 2	103.91	185.74	452.3	1,163.48	3,015.64	7,821.43
SubAssembly Supplier 3	51.95	92.87	226.15	581.74	1,507.82	3,910.72



SCENARIO 2 : ASSUME UNKNOWN SALES INCREASE EACH WEEK

Strategies to Ensure Order Fulfillment without Excess Stock



1. Implement a Real-Time Demand Tracking System

Instead of relying on past order patterns, use real-time sales data from retailers and RDCs to adjust production and inventory dynamically. This allows manufacturers to scale production gradually rather than react to outdated forecasts.

Benefit: Reduces the risk of overproducing based on incorrect predictions.

Trade-off: Requires investment in data analytics and system upgrades to enable real-time decision-making.



2. Reduce Lead Times for Faster Adaptation

Shorten the Inventory Adjustment Time (4 weeks) to 2-3 weeks where possible, allowing supply chain entities to react faster to demand changes. This can be done by:

- Enhancing supplier coordination to ensure raw materials arrive faster.
- Using agile manufacturing methods to reduce production cycle time.
- Optimizing transportation routes to speed up distribution.

Benefit: Enables quicker response to demand changes, reducing stock imbalances.

Trade-off: May require higher logistics costs and supplier flexibility, which could increase short-term expenses.



3. Use a Flexible Safety Stock Strategy

Adjust buffer levels dynamically based on demand trends instead of maintaining a fixed 1-week safety stock.

Retailers and RDCs should increase safety stock moderately during initial fluctuations.

Manufacturers should increase production in smaller increments to avoid large inventory spikes.

Benefit: Ensures customer orders are met without excessive stock buildup.

Trade-off: Risk of temporary shortages if demand grows faster than anticipated.



4. Shift to a Pull-Based Replenishment Model

Adopt a just-in-time (JIT) approach, where inventory replenishment is based on actual sales rather than speculative demand forecasting. Limit how much upstream suppliers can increase production per cycle to prevent overstocking.

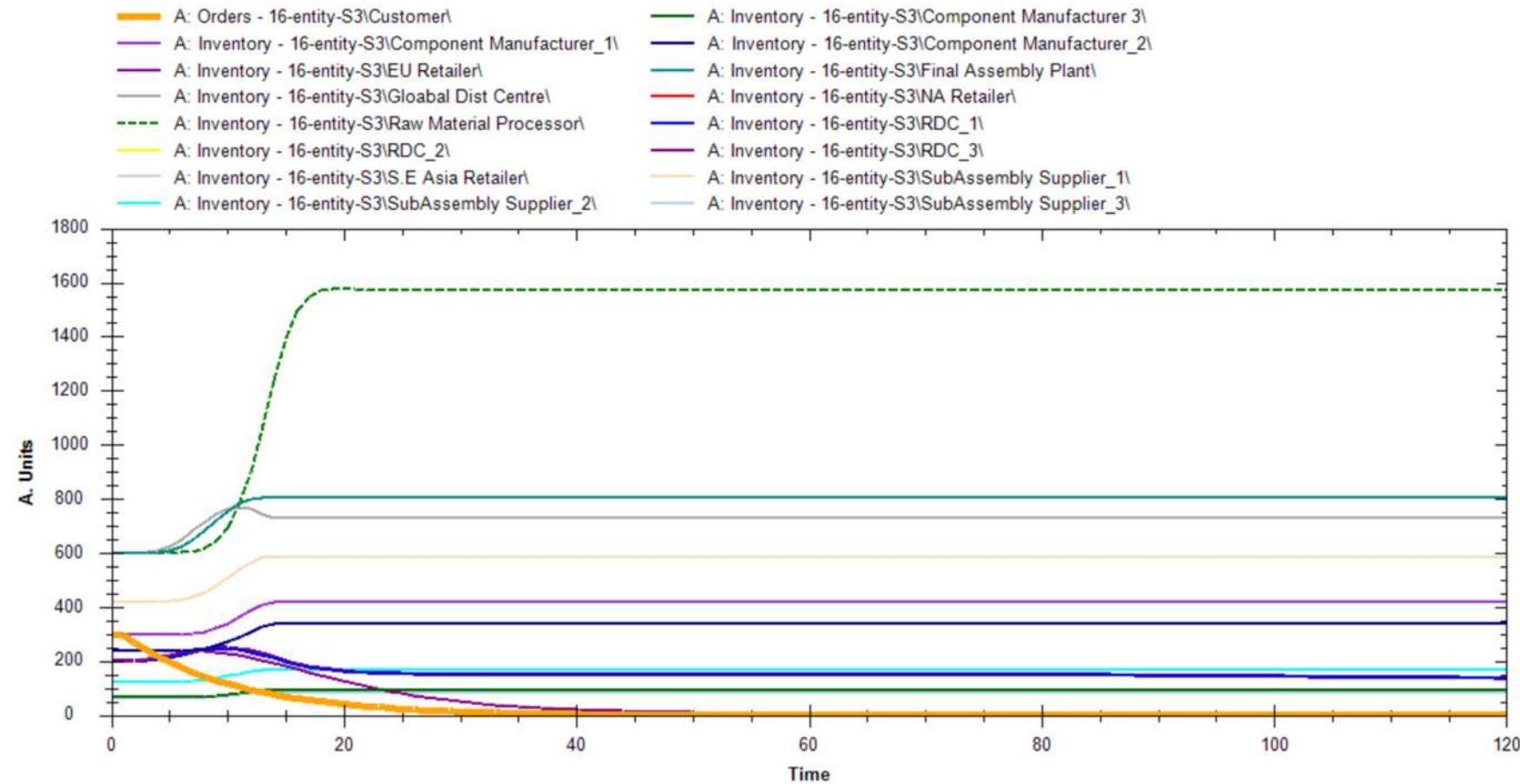
Adjust order quantities based on a rolling 4-week average demand rather than reacting to short-term spikes.

Benefit: Prevents overproduction and reduces storage costs.

Trade-off: If demand surges unexpectedly, some supply chain entities may experience stockouts before production catches up.



SCENARIO 3 : ASSUME UNKOWN SALES DECREASE EACH WEEK



Initial order volume: 300 units per week
Starting Week 2: Demand decreased by 10% each week
Growth Pattern: Compounding increase (week-over-week)
Duration: Continues until Week 52

WEEK	10	20	30	40	50	60
Component Manufacturer 1	297	339.65	420.22	420.22	420.22	420.22
Component Manufacturer 2	237	271.03	335.33	335.33	335.33	335.33
Component Manufacturer 3	66	75.48	93.38	93.38	93.38	93.38
Customer	300	116	41	14	5	2
European Retailer	198	229.4	125.98	50.4	20.37	9.81
Final Assembly Plant	600	754.52	803.61	803.61	803.61	803.61
Global Dist. Centre	600	761.79	727.6	727.6	727.6	727.6
N.America Retailer	198	229.4	125.98	50.4	20.37	9.81
Raw Material Processor	600	695.17	1,576.54	1,568.14	1,568.18	1,568.18
RDC 1	198	244.11	161	149.14	149.14	149.14
RDC 2	198	244.11	161	149.14	149.14	149.14
RDC 3	204	251.51	165.88	153.66	153.66	153.66
S.E. Asia Retailer	204	236.35	129.8	51.92	20.98	10.1
SubAssembly Supplier 1	420	507.17	583.57	583.57	583.57	583.57
SubAssembly Supplier 2	120	144.91	166.73	166.73	166.73	166.73
SubAssembly Supplier 3	60	72.45	83.37	83.37	83.37	83.37

Strategies to Ensure Order Fulfillment without Excess Stock

1. Implement Real-Time Inventory Monitoring

Track inventory velocity and align replenishment frequency to avoid over-ordering.

Trade-off: Requires agile systems and tight execution discipline.

2. Scale Down Production Responsively

Introduce dynamic production scheduling (e.g., reduce batch sizes) to match shrinking demand.

Trade-off: Potential underutilization of production resources.

3. Slow Down Procurement Cycles

Delay raw material/component purchases by adjusting the average order rate and reordering point.

Trade-off: Could increase lead times for re-ramping if demand recovers.

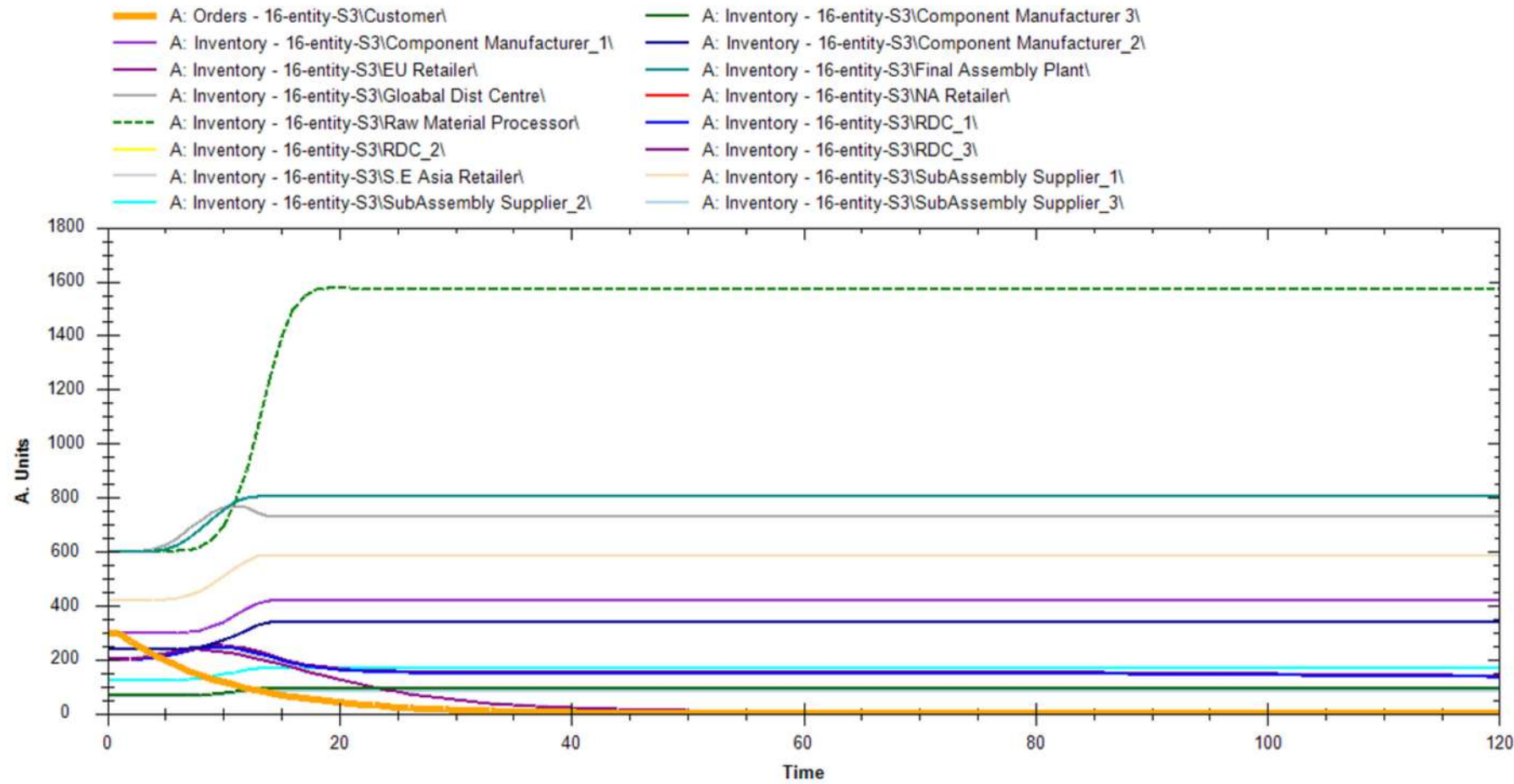
4. Use Targeted Inventory Burn-Off Plans

Focus on draining excess inventory in specific nodes (e.g., subassembly suppliers or RDCs) before restocking.

Trade-off: Risk of localized stockouts if demand stabilizes unexpectedly.



SCENARIO 4 : DEALING WITH ONGOING DEMAND UNCERTAINTY



Initial customer order: 300 units per week
Starting Week 10-55: Demand fluctuates until Week 55
New order rate: 1378 units per week
Duration: Sustained until Week 120

Strategies to Handle Demand Uncertainty

1. Adopt Time-Phased Safety Stock Strategies

Use variable safety stock that scales with demand volatility; hold higher stock during uncertain windows.

Trade-off: Increased inventory costs, but ensures service levels.

2. Decentralized Inventory Holding

Position buffer inventory at RDCs and Final Assembly plants rather than only upstream.

Trade-off: Higher storage and transportation costs, but enables agility.

3. Use Dual Sourcing or Backup Capacity

Enable quick switches between suppliers or geographies when sudden changes occur.

Trade-off: More complex supplier management and potentially higher procurement cost.

4. Simulate Demand Scenarios Regularly

Run digital twin simulations to stress-test different demand patterns and prepare contingency plans.

Trade-off: Needs analytics capabilities and scenario planning teams.

WEEK	10	20	30	40	50	60
Component Manufacturer 1	297	260.97	419.2	930.95	235.56	719.92
Component Manufacturer 2	237	208.24	334.51	742.88	187.97	574.48
Component Manufacturer 3	66	57.99	93.16	206.88	52.35	159.98
Customer	360	561	348	604	1,080.00	1,378.00
European Retailer	198	275.38	448.5	239.77	390.82	677.88
Final Assembly Plant	600	627.99	942.05	1,376.61	792.19	1,716.72
Global Dist. Centre	600	722.98	1,073.28	990.11	955.72	1,832.52
N.America Retailer	198	275.38	448.5	239.77	390.82	677.88
Raw Material Processor	600	404.24	725.85	3,212.43	3,406.05	815.3
RDC 1	198	271.52	410.67	257.09	357.32	643.02
RDC 2	198	271.52	410.67	257.09	357.32	643.02
RDC 3	204	279.75	423.11	264.88	368.14	662.51
S.E. Asia Retailer	204	283.72	462.09	247.03	402.67	698.42
SubAssembly Supplier 1	420	397.85	612.46	1,252.40	410.59	1,104.13
SubAssembly Supplier 2	120	113.67	174.99	357.83	117.31	315.47
SubAssembly Supplier 3	60	56.84	87.49	178.91	58.66	157.73



SCENARIO 5: DEALING WITH SEASONAL ADJUSTMENT



Background: Zara's Seasonal Supply Chain in Asia

Zara operates a fast-fashion supply chain in the Asia-Pacific region, known for frequent product launches, short life cycles, and high responsiveness to seasonal trends. In this simulation, the 16-entity digital twin supply chain is used to model Zara's regional operations, covering the journey from raw materials to customer-facing retailers.

The supply chain consists of raw material processors, component manufacturers (producing fabrics, zippers, and trims), subassembly suppliers (handling dyeing, cutting, and stitching), a final assembly plant (producing finished garments), and distribution centers (RDCs) serving three key markets. Customer demand represents the combined orders from Zara stores across Southeast Asia.

The scenario includes three key seasonal phases:

Week 1–15: Stable demand at 300 units/week

Week 16–25: Spring/Summer promotional season – demand surges to 420 units/week

Week 26–30: Transition period – demand dips to 240 units/week

Week 31–45: Fall/Winter collection – demand peaks at 450 units/week

Week 46–120: End-of-season sales – demand stabilizes at 300 units/week

In response to different seasonal collections, Zara modifies its internal assembly structure. For example, **the Final Assembly Plant adjusts its sourcing ratio: from 70% SubAssembly 1 / 20% Sub 2 / 10% Sub 3 (Weeks 1–30) to 50% Sub 1 / 40% Sub 2 / 10% Sub 3 (Weeks 31–120).**

This reflects changing styles, materials, and sourcing needs for different fashion lines.



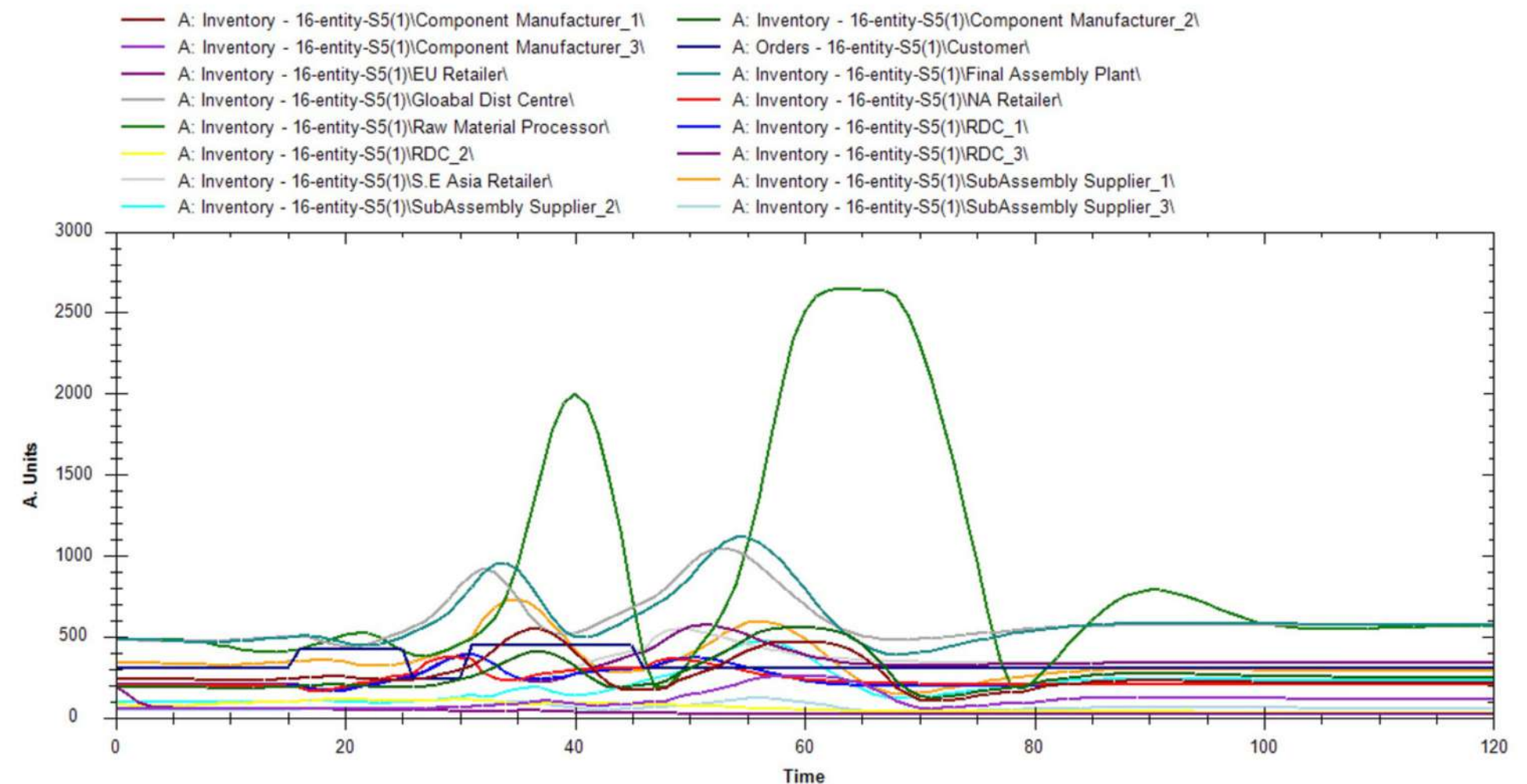
SCENARIO 5: DEALING WITH SEASONAL ADJUSTMENT

How can Upstream Suppliers Respond Efficiently?

- **Seasonal Inventory Buffering:** Component manufacturers and subassembly suppliers should build advance inventory buffers during stable demand periods (e.g., Weeks 10–15 and Weeks 26–30) in preparation for known seasonal peaks (e.g., Weeks 16–25 and 31–45). This allows upstream suppliers to fulfill sudden order spikes without overreacting and avoids panic-driven production increases.
- **Flexible Production Windows:** Upstream suppliers can establish flexible production windows aligned with seasonal calendars. For example, manufacturers may temporarily scale up production capacity by 20–30% during Weeks 14–16 and 29–31, coinciding with the beginning of promotional seasons. This ensures timely fulfillment without continuous overproduction.
- **Shorter Adjustment & Lead Times:** Reducing the inventory adjustment time and order processing lead time from 4 weeks to 2–3 weeks enables faster reaction to changing downstream needs. This responsiveness minimizes stockouts and avoids excessive buildup of unsold goods at subassembly or component levels.
- **Collaborative Forecast Sharing:** Component and subassembly suppliers should be integrated into a shared forecast system, enabling them to see upcoming promotional windows or collection changes. With better visibility into Zara's retail planning, they can adjust production more accurately and reduce guesswork.
- **Dynamic Sourcing Strategy:** If certain component suppliers become bottlenecks (e.g., Component Manufacturer 1), Zara can work with secondary sources or local alternatives for specific fabrics or trims. This dual-sourcing model helps maintain continuity without overstocking from a single supplier.

Initial customer order: 300 units per week
Starting Week 2: Demand fluctuates seasonally
Duration: Continues until Week 120
Final Assembly Plant Order Distribution:

- 70%, 20%, 10% during Week 1-30,
- 50%, 40%, 10% during Week 31-120





SCENARIO 5: RECOMMENDATIONS TO ZARA'S CHIEF SUPPLY CHAIN OFFICER



Short-Term Recommendations (1–2 Years)

Visibility & Agility

1. Enable store-level sales tracking and upstream data sharing through collaborative dashboards to align all tiers of the supply chain, especially during promotional events.
2. Position safety stock at RDCs and final assembly nodes, and utilize nearshore suppliers or air freight for flexible, short-term replenishment during demand spikes.
3. Integrate promotional calendars with supply planning and simulate campaign impacts using the digital twin to ensure readiness across suppliers and production.

Goal: Faster response, reduced upstream inventory buildup, and improved customer fulfillment.



Medium-Term Recommendations (2–3 Years)

Flexibility & Supplier Agility

1. Adopt modular design strategies and delayed differentiation to produce neutral garments that can be quickly finalized based on near-term demand signals.
2. Implement tier-specific safety stock rules with low upstream inventory and buffered downstream stock, using real-time demand data to auto-adjust levels.
3. Expand and diversify the supplier base across nearshore regions, and establish rolling capacity agreements to enable rapid production scaling as needed.

Goal: Increase responsiveness while minimizing the risk of overproduction and supplier strain.



Long-Term Recommendations (3–5 Years)

Predictive, Digital & Sustainable Supply Chain

1. Scale the digital twin to connect real-time retail, production, and supplier data, enabling scenario planning for market shifts, supply risks, and regulatory changes.
2. Leverage machine learning models to forecast demand using social trends, weather patterns, and POS data, optimizing inventory through predictive analytics.
3. Invest in reverse logistics and recycling programs for unsold or returned inventory, while reducing carbon emissions through smarter transport and sourcing.

Goal: Build a resilient, data-driven, and environmentally responsible supply chain.